

МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО
ОБРАЗОВАНИЯ РОССИЙСКОЙ ФЕДЕРАЦИИ

ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ БЮДЖЕТНОЕ
ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ

**«БЕЛГОРОДСКИЙ ГОСУДАРСТВЕННЫЙ
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Лабораторная работа №2

по дисциплине: Тестирование программных систем
тема: «SoapUI как инструмент эмуляции сервисов»

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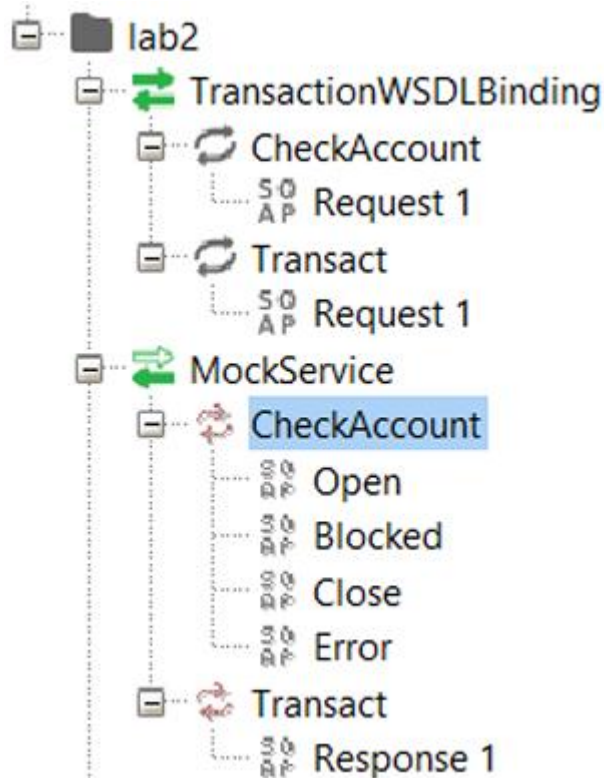
Цель работы: изучить принципы работы со средством эмуляции работы веб сервисов.

Для операции CheckAccount реализовать:

- Возвращать Close, Blocked, Open при получении определенных значений AccountNumber (значения выбрать самостоятельно).
- Для остальных значений Account Number возвращать ошибочный ответ.
- В AccountNumber всегда возвращать входное значение.
- Name и Surname в запросе указывать свое.

Ход работы:

Структура проекта:



Request 1:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:wsc="http://bank/schema/WSCommonTypes">
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountRequest>
      <wsc:AccountNumber>3000</wsc:AccountNumber>
      <wsc:Name>Андрей</wsc:Name>
      <wsc:Surname>Дмитриев</wsc:Surname>
      <wsc:Passport>12 34 567890</wsc:Passport>
    </wsc:CheckAccountRequest>
  </soapenv:Body>
</soapenv:Envelope>
```

Пример одного из респонсов (Open):

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:wsc="http://bank/schema/WSCCommonTypes">
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountResponse>
      <wsc:AccountNumber>${accountNumber}</wsc:AccountNumber>
      <wsc:Status>Open</wsc:Status>
    </wsc:CheckAccountResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

– также описаны остальные ответы

Скрипт:

```
def groovyUtils = new com.eviware.soapui.support.GroovyUtils(context)
def holder = groovyUtils.getXmlHolder(mockRequest.requestContent)
holder.declareNamespace("wsc", "http://bank/schema/WSCCommonTypes")

def accountNumber =
holder.getNodeValue("//wsc:CheckAccountRequest/wsc:AccountNumber")

log.info("Received AccountNumber: " + accountNumber)

context.accountNumber = accountNumber

if (accountNumber == "1000") {
  log.info("Returning Open response")
  return "Open"
} else if (accountNumber == "2000") {
  log.info("Returning Blocked response")
  return "Blocked"
} else if (accountNumber == "3000") {
  log.info("Returning Close response")
  return "Close"
} else {
  log.info("Returning Error response for account: " + accountNumber)
  return "Error"
}
```

Запрос 1000 – Open:

The screenshot shows a web browser window with the address bar displaying `http://localhost:8088/mock`. The page content is divided into two main sections: a raw XML view on the left and a structured view on the right.

Raw XML View (Left):

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:wsc="http://b...">
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountRequest>
      <wsc:AccountNumber>1000</wsc:AccountNumber>
      <wsc:Name>Александр</wsc:Name>
      <wsc:Surname>Чечин</wsc:Surname>
      <wsc:Passport>11 11 11111</wsc:Passport>
    </wsc:CheckAccountRequest>
  </soapenv:Body>
</soapenv:Envelope>
```

Structured View (Right):

The structured view shows the response details. The status is `HTTP/1.1 200 OK`. The server is `Jetty(6.1.26)`. The content type is `text/xml; charset=utf-8`.

Header	Value
#status#	HTTP/1.1 200 OK
Server	Jetty(6.1.26)
Content-Type	text/xml; charset=utf-8

The bottom of the browser window shows tabs for `Headers (3)`, `Attachments (0)`, `SSL Info`, `WSS (0)`, and `JMS (0)`.

Запрос 2000 – Blocked:

The screenshot shows a web browser window with the address bar displaying `http://localhost:8088/mock`. The page content is divided into two main sections: a raw XML view on the left and a structured view on the right.

Raw XML View (Left):

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:wsc="http://b...">
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountRequest>
      <wsc:AccountNumber>2000</wsc:AccountNumber>
      <wsc:Name>Александр</wsc:Name>
      <wsc:Surname>Чечин</wsc:Surname>
      <wsc:Passport>11 11 11111</wsc:Passport>
    </wsc:CheckAccountRequest>
  </soapenv:Body>
</soapenv:Envelope>
```

Structured View (Right):

The structured view shows the response details. The status is `HTTP/1.1 200 OK`. The server is `Jetty(6.1.26)`. The content type is `text/xml; charset=utf-8`.

Header	Value
#status#	HTTP/1.1 200 OK
Server	Jetty(6.1.26)
Content-Type	text/xml; charset=utf-8

The bottom of the browser window shows tabs for `Headers (3)`, `Attachments (0)`, `SSL Info`, `WSS (0)`, and `JMS (0)`.

Запрос 3000 –Close:

The screenshot displays a SOAP request and response in SoapUI. The request is a `CheckAccountRequest` with account number 3000 and status 'Close'. The response is a `CheckAccountResponse` with the same account number and status 'Close'. The status bar indicates a successful response (200 OK).

Raw XML

Request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountRequest>
      <wsc:AccountNumber>3000</wsc:AccountNumber>
      <wsc:Name>Александр</wsc:Name>
      <wsc:Surname>Чечин</wsc:Surname>
      <wsc:Passport>11 11 11111</wsc:Passport>
    </wsc:CheckAccountRequest>
  </soapenv:Body>
</soapenv:Envelope>
```

Response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:wsc="http://b...>
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountResponse>
      <wsc:AccountNumber>3000</wsc:AccountNumber>
      <wsc:Status>Close</wsc:Status>
    </wsc:CheckAccountResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

Headers:

Header	Value
#status#	HTTP/1.1 200 OK
Server	Jetty(6.1.26)
Content-Type	text/xml; charset=utf-8

Headers (3) Attachments (0) SSL Info WSS (0) JMS (0)

Ошибочный запрос:

The screenshot displays an erroneous SOAP request and response in SoapUI. The request is a `CheckAccountRequest` with account number 999. The response is a `CheckAccountResponse` with account number 999 and an error message 'Account not found'. The status bar indicates a successful response (200 OK).

Raw XML

Request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountRequest>
      <wsc:AccountNumber>999</wsc:AccountNumber>
      <wsc:Name>Александр</wsc:Name>
      <wsc:Surname>Чечин</wsc:Surname>
      <wsc:Passport>11 11 11111</wsc:Passport>
    </wsc:CheckAccountRequest>
  </soapenv:Body>
</soapenv:Envelope>
```

Response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:wsc="http://b...>
  <soapenv:Header/>
  <soapenv:Body>
    <wsc:CheckAccountResponse>
      <wsc:AccountNumber>999</wsc:AccountNumber>
      <wsc:Error>Account not found</wsc:Error>
    </wsc:CheckAccountResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

Headers:

Header	Value
#status#	HTTP/1.1 200 OK
Server	Jetty(6.1.26)
Content-Type	text/xml; charset=utf-8

Headers (3) Attachments (0) SSL Info WSS (0) JMS (0)

Вывод: В ходе лабораторной работы изучены принципы работы со средством эмуляции работы веб-сервисов.